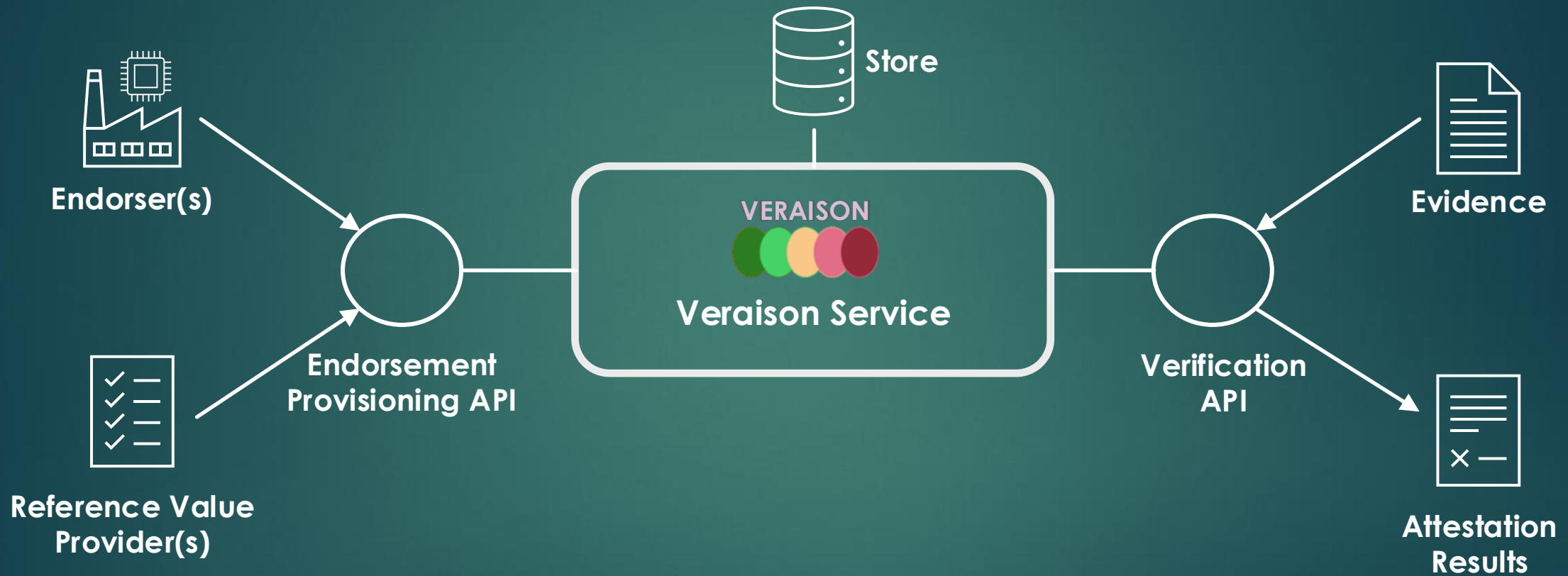


Endorsement Distribution with **Veraison** and **CoSERV**

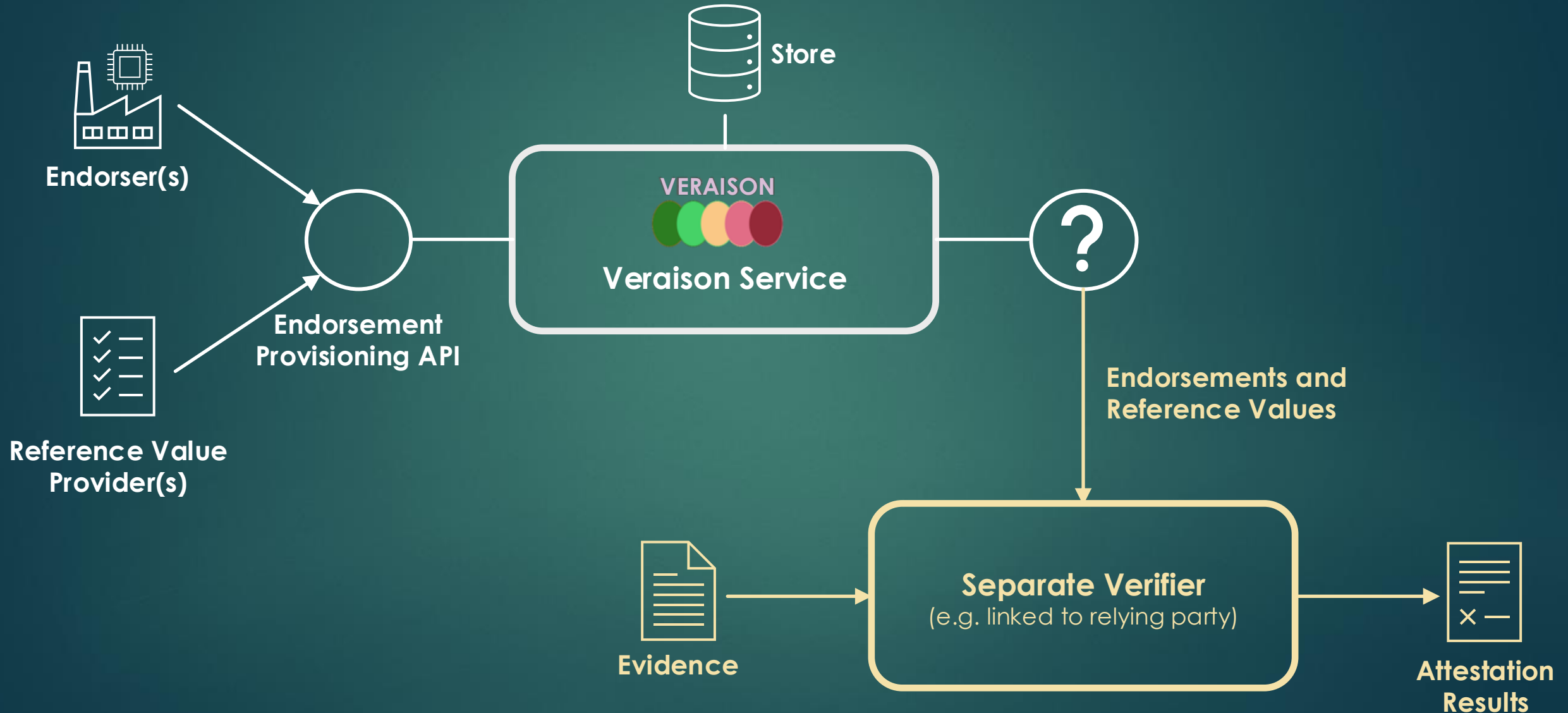
CCC ATTESTATION SIG, JUNE 2026

PAUL HOWARD, ARM

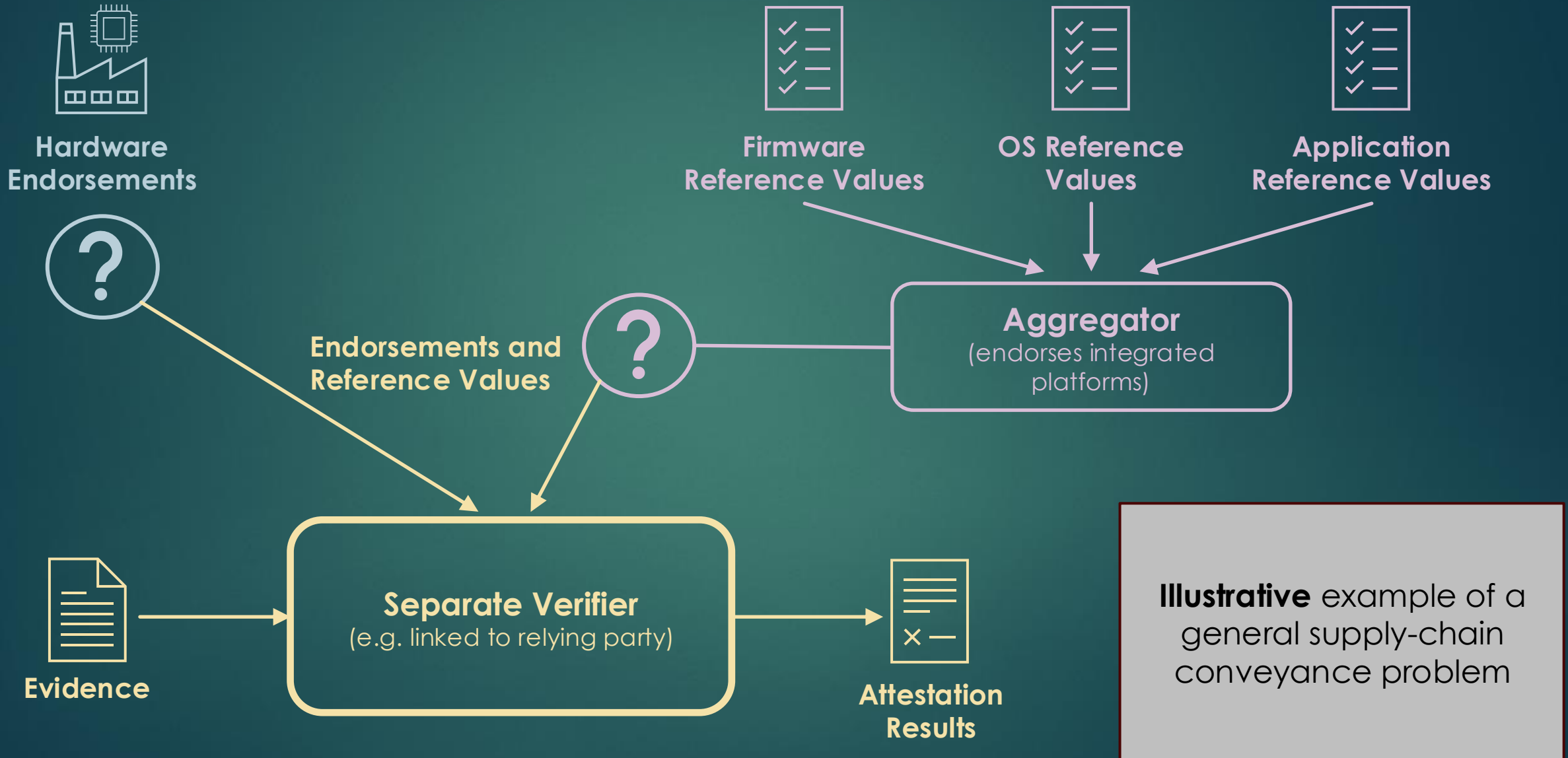
Veraison Verification Service Model



Distributed Verification Model



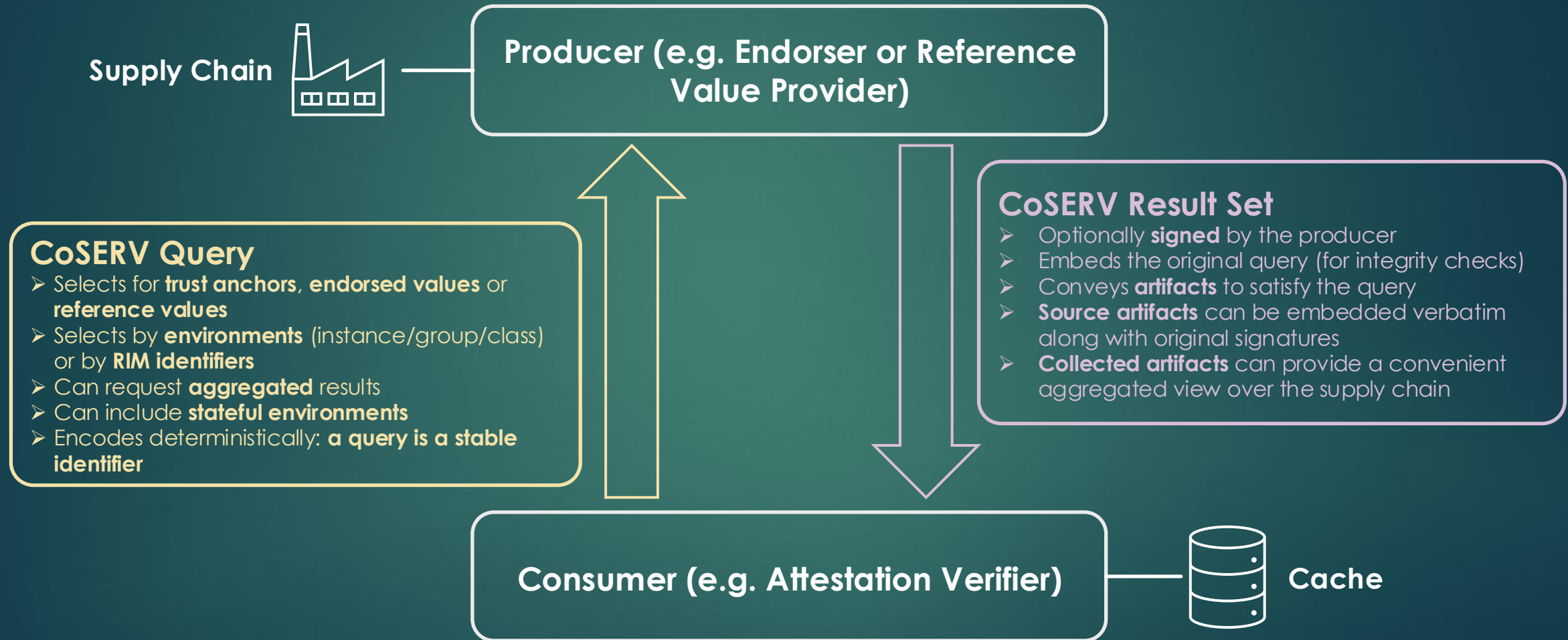
Harmonized, Flexible Distribution



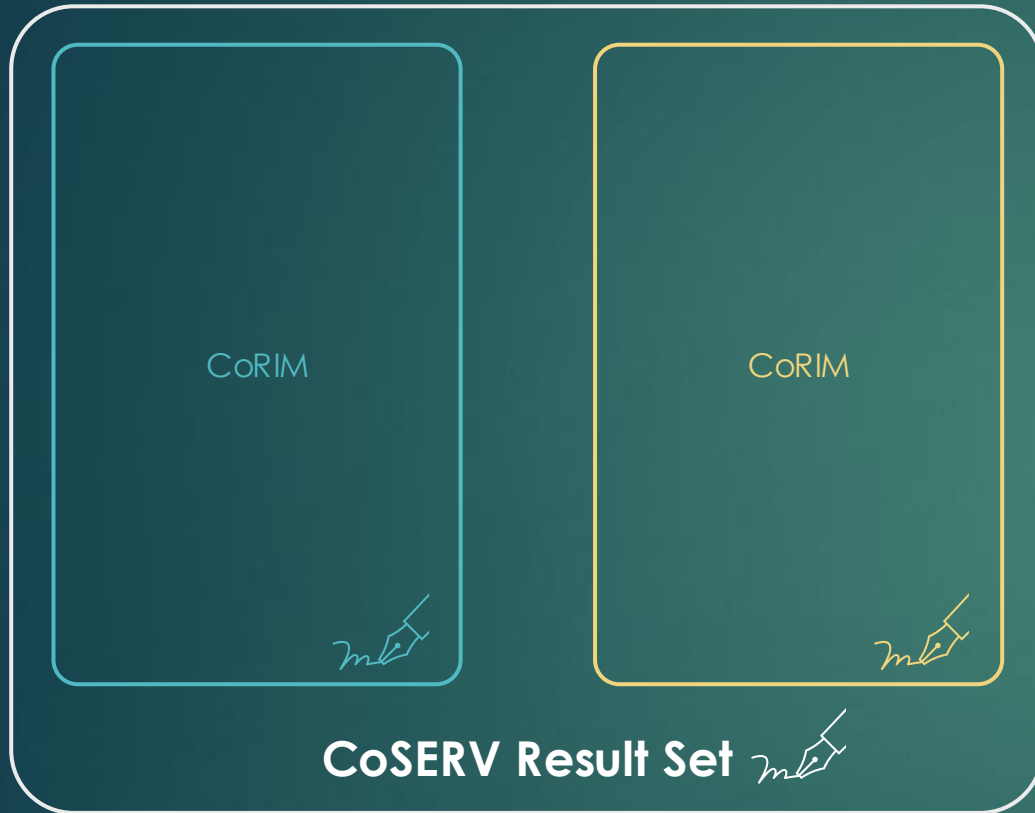
Introducing CoSERV

- ▶ **CoSERV** = **C**oncise **S**elector for **E**ndorsements and **R**eference **V**alues
- ▶ Active (adopted) [IETF draft](#) within the RATS Working Group
- ▶ Closely related to CoRIM; directly inherits CoRIM data model
- ▶ Fulfills a function that CoRIM lacks: the ability to query for the required RIMs
- ▶ Encoded in CBOR for efficiency
- ▶ Minimal query language, specialised for endorsement and reference value artifacts
- ▶ Could operate over different transports
- ▶ HTTP REST API included in the specification
 - ▶ Simple request-response protocol
 - ▶ Cache-friendly: CoSERV queries are stable identifiers

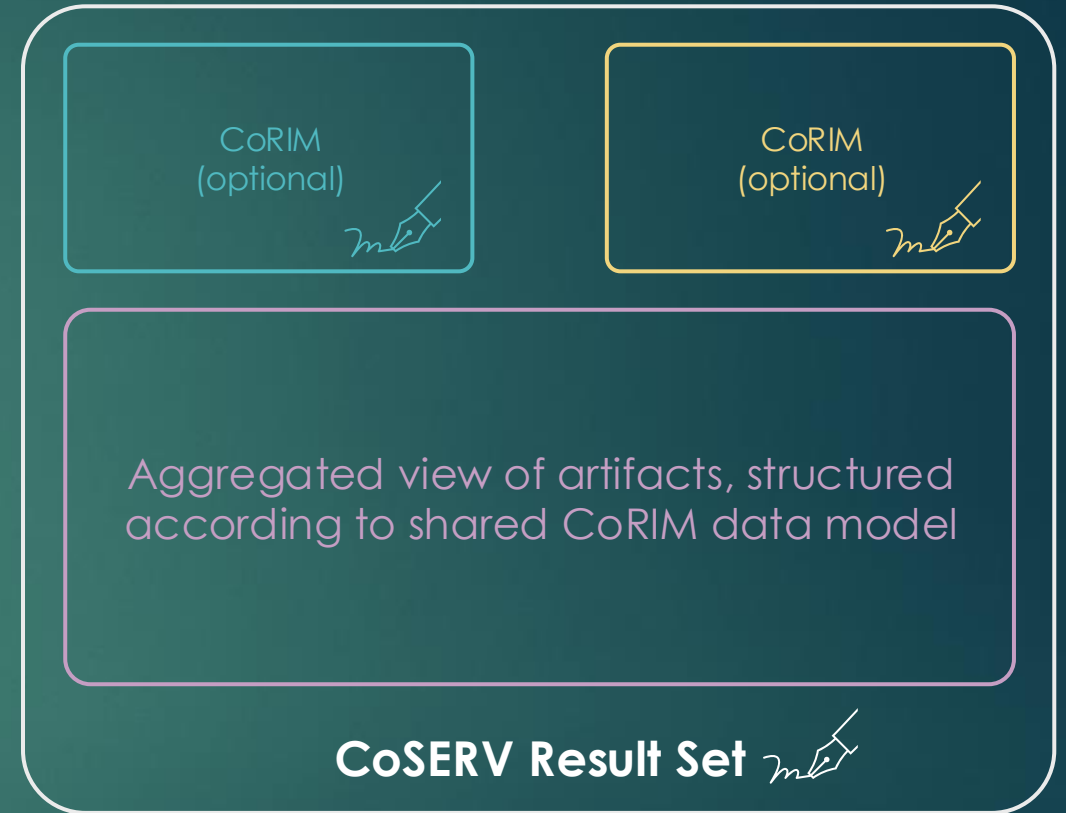
A CoSERV Transaction



CoSERV Result Sets



At its simplest, the CoSERV result is a wrapper around one or more CoRIM (or equivalent) artifacts



CoSERV providers can aggregate results from multiple supply chain sources into a single view that is easy for the verifier to consume

Why Not Just Static RIM Locators?

- ▶ With the HTTP API binding, a CoSERV query **is** just a URL!

```
GET https://veraison.test.linaro.org:11443/endorsement-distribution/v1/coserv/<base64url-string>
```

- ▶ CoSERV query URLs are stable/cacheable using off-the-shelf HTTP middleware
- ▶ Server can apply intelligence about cache control and refresh cadence (which might be more frequent than individual CoRIM lifetimes)
- ▶ Can conveniently package multiple RIMs in a single transaction
- ▶ Can handle stateful environments (yielding different results)
- ▶ Hits the “sweet spot” between dynamic capability when needed, but with reduced loads on the server due to caching
- ▶ Some existing endorsement services need dynamic behaviour, so any proposed standard needs to cater for that to invite broad adoption

Veraison is the RATS Proving Ground



The **IETF RATS WG** is the ideal place to define a harmonized and flexible conveyance protocol for endorsements and reference values

The **Veraison** project is the ideal place to show how it can be implemented and used in practice

Veraison CoSERV Support: Libraries

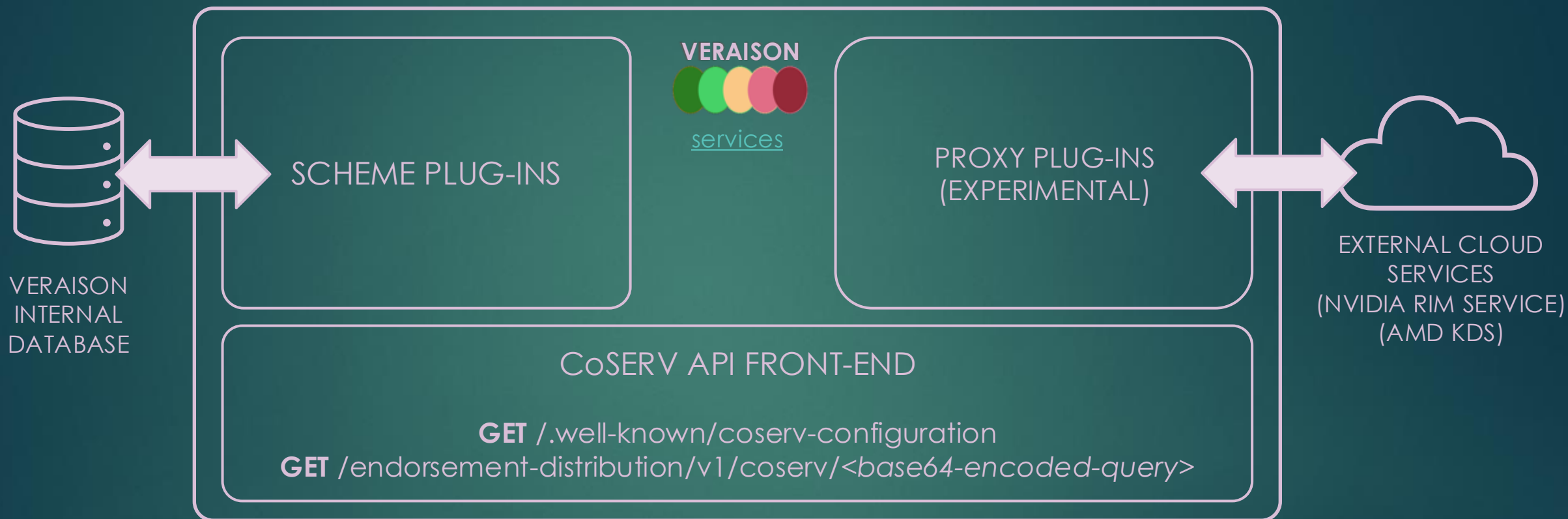


- ✓ Data model and CBOR serialization/de-serialization, plus COSE sign/verify support:
<https://github.com/veraison/corim>



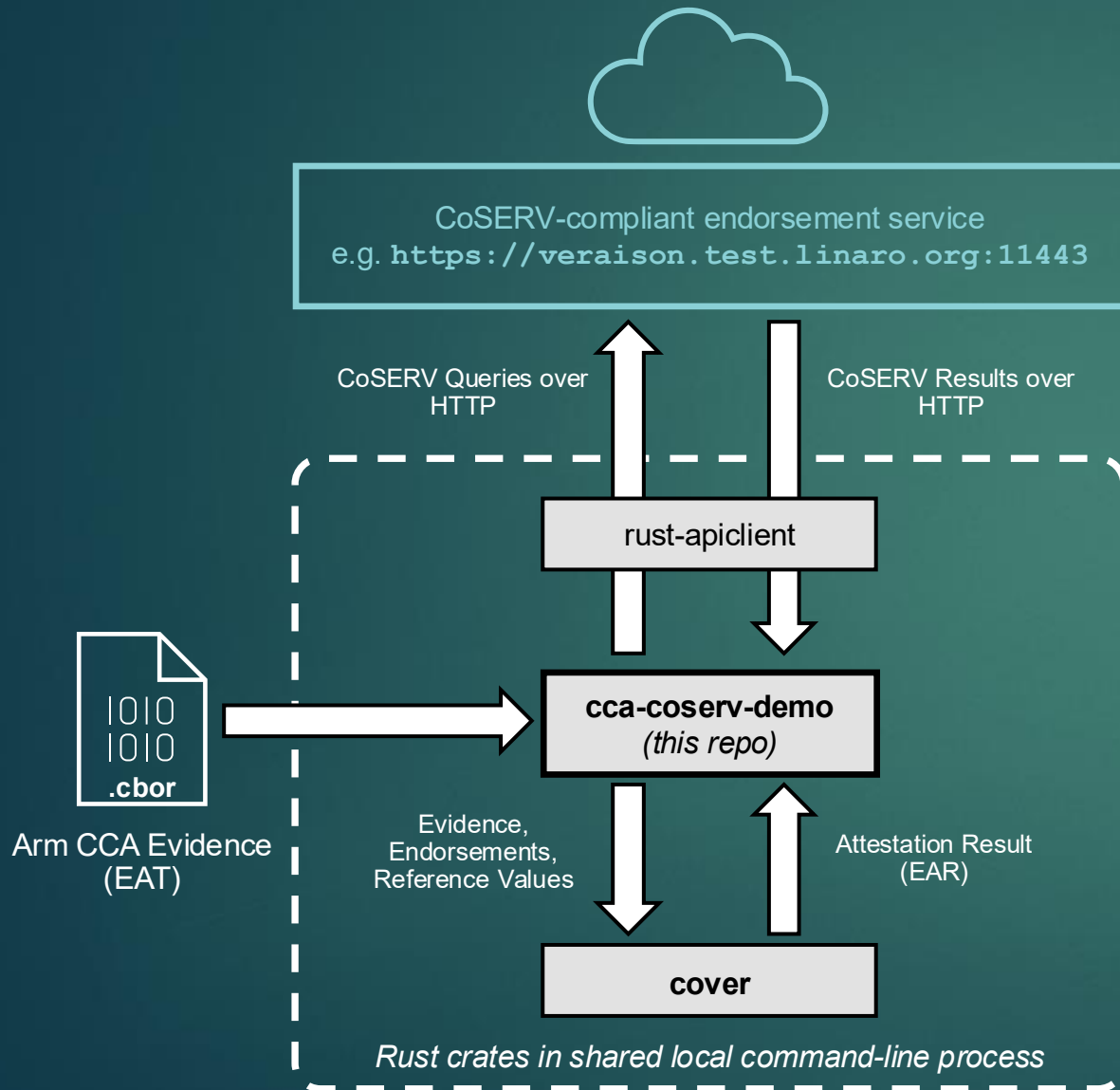
- ✓ Data model and CBOR serialization/de-serialization, plus COSE sign/verify support:
<https://github.com/veraison/coserv-rs>
- ✓ Client-side support for running and caching CoSERV queries:
<https://github.com/veraison/rust-apiclient>

Veraison CoSERV Support: Services



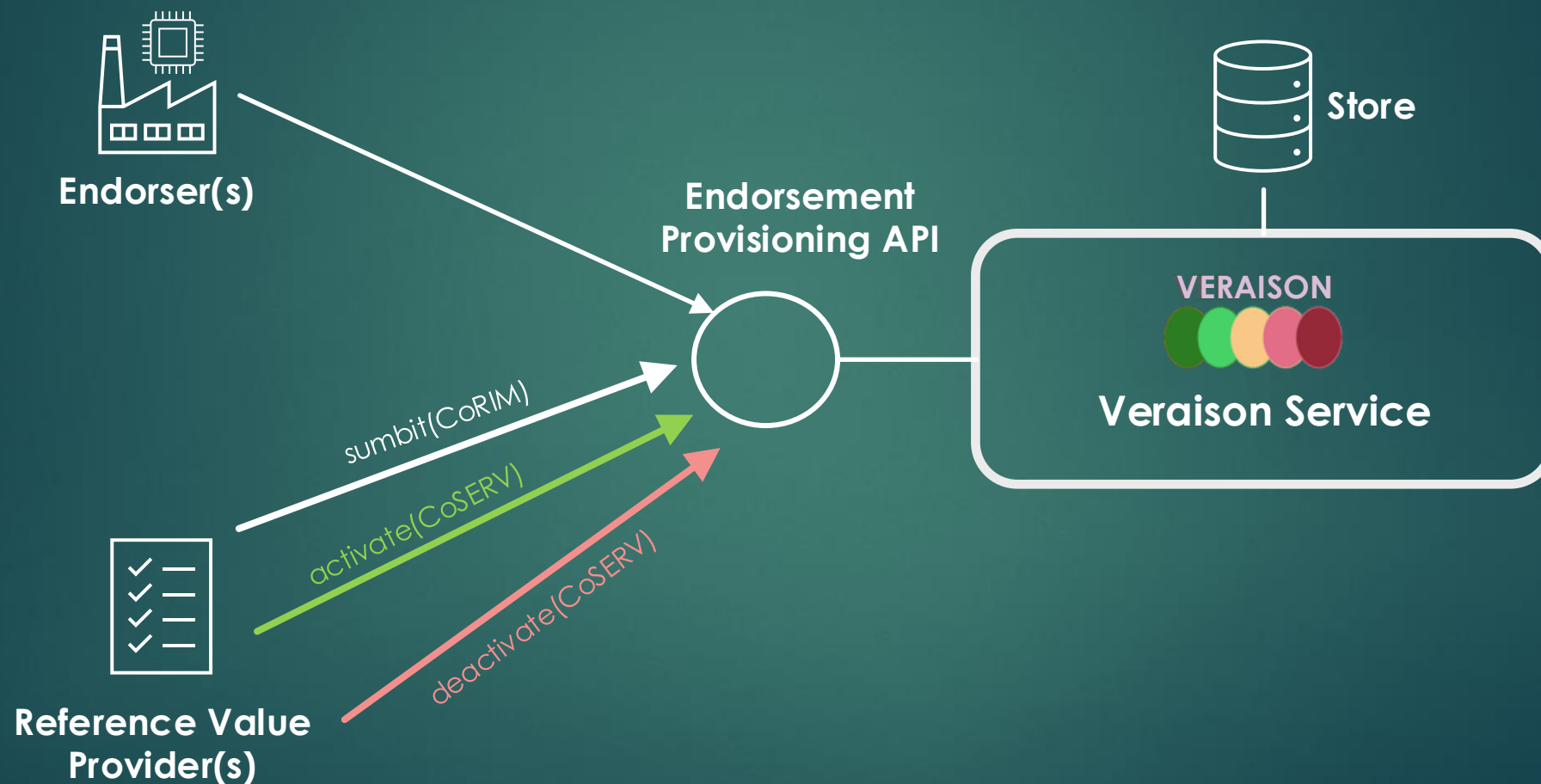
Example live deployment available for Arm CCA pre-silicon environments:
<https://veraison.test.linaro.org:11443>

End-to-End Demonstrator (Arm CCA)



- ▶ Fully encapsulated demo at: <https://github.com/veraison/cca-coserv-demo>
- ▶ Presented at IETF-125 Hackathon
- ▶ Supports verification and appraisal of Arm CCA evidence, but would extend to other schemes
- ▶ Showcases server-side support in Veraison services
- ▶ Showcases Rust client-side support, including caching (can be run “unplugged” while cached results are valid)
- ▶ Uses the new **CoRIM verifier** from the Veraison project: <https://github.com/veraison/cover>
- ▶ Demonstrates CoRIM/CoSERV compatibility and combined value
- ▶ All functionality in mainline branches

New Directions: CoSERV for Lifecycle Control



Special Thanks



Thomas Fossati and the team at **Linaro**



Shefali Kamal, Dhanus Lal and the team at **Fujitsu**



Sergei Trofimov, Yogesh Deshpande and the team at **Arm**



Carl Wallace, Giri Mandyam, Henk Birkholz, Ding Ma

Resources



- ▶ **Draft:** <https://datatracker.ietf.org/doc/draft-ietf-rats-coserv/05/>



- ▶ **Draft repo** (for issues and PRs): <https://github.com/ietf-rats-wg/draft-ietf-rats-coserv>



- ▶ **End-to-end demo:** <https://github.com/veraison/cca-coserv-demo>

Questions?

Thank You!